

Lorenzo Esposito

PhD candidate in Psychology and Cognitive Sciences

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RESEARCH INTERESTS

Mathematics learning and math anxiety; individual and sex/gender differences in cognition and achievement; intelligence.

EDUCATION

PhD in Social Sciences, curriculum Psychology and Cognitive Sciences

Nov 2023 – present

University of Genoa, XXXIX cycle

Supervisor: Prof. David Giofrè

MSc in Cognitive Neuroscience and Clinical Neuropsychology

Sep 2019 – Jul 2021

University of Padua · 110/110 *cum laude*

Thesis: *Representation of small vs large numerical quantities and the role of number notation:*

An electrophysiological study

Supervisor: Prof. Marco Zorzi

BSc in Cognitive and Psychobiological Psychology

Oct 2016 – Jul 2019

University of Padua · 110/110 *cum laude*

Thesis: *Discrimination of simple geometric shapes in zebrafish larvae*

Supervisor: Prof. Angelo Bisazza

Professional Qualification as a Psychologist

Jul 2023

University of Genoa

RESEARCH EXPERIENCE

Visiting PhD student

Jan – Aug 2026

University of Ulm, Germany

Supervisor: Prof. Oliver Wilhelm

Visiting PhD student

Jul – Aug 2025

University of Amsterdam, the Netherlands

Supervisor: Prof. Dylan Molenaar

Research assistant

Dec 2022 – Oct 2023

Department of Developmental Psychology and Socialisation, University of Padua

Post-graduate intern

Dec 2021 – Nov 2022

Department of Developmental Psychology and Socialisation, University of Padua

Including Apr – Nov 2022 at the Lab.D.A. clinical centre, Padua

Pre-graduate intern

Feb – May 2021

Department of General Psychology, University of Padua

PUBLICATIONS

Peer-reviewed articles

1. **Esposito, L.**, & Giofrè, D. (2026). Investigating the impact of attitudes and learning environment on math achievement: Insights from the TIMSS data. *Personality and Individual Differences*, 264, 114046. DOI
2. **Esposito, L.**, Molenaar, D., & Giofrè, D. (2026). Testing cross-group comparability of INVALSI scores using moderated factor analysis. *Intelligence*, 118, 102042. DOI
3. Caviola, S., Giofrè, D., Toffalini, E., **Esposito, L.**, Tonizzi, I., & Geary, D. C. (2026). Sex differences in mathematics: A meta-analytic review across content, grades, and geographic regions. *Educational Psychology Review*, 38, 58. DOI
4. **Esposito, L.**, Tonizzi, I., Usai, M. C., & Giofrè, D. (2026). The impact of math attitudes and gender in future school choice: A longitudinal study among Italian students. *Journal of Intelligence*, 14(3), 38. DOI
5. **Esposito, L.**, Toffalini, E., & Giofrè, D. (2026). Sex differences in mathematics and reading across Italy: The role of socioeconomic factors. *Intelligence & Cognitive Abilities*, 2(1). DOI
6. Ikeda, Y., **Esposito, L.**, Kita, Y., Oi, Y., Takagi, R., Suzuki, K., Mammarella, I. C., Caviola, S., Lanfranchi, S., Pulina, F., & Giofrè, D. (2026). Cognitive and affective-emotional factors in math achievement: The mediating role of intelligence. *Journal of Intelligence*, 14(2), 25. DOI
7. **Esposito, L.**, Giofrè, D., Toffalini, E., & Geary, D. C. (2025). Sex differences in variability: Evidence from math and reading assessment in Italy. *Personality and Individual Differences*, 247, 113410. DOI
8. **Esposito, L.**, Tonizzi, I., Usai, M. C., & Giofrè, D. (2025). Understanding the role of cognitive abilities and math anxiety in adolescent math achievement. *Journal of Intelligence*, 13(4), 44. DOI
9. **Esposito, L.**, & Giofrè, D. (2025). Sex differences in intelligence on the WISC: A meta-analysis on children with specific learning disabilities. *Journal of Intelligence*, 13(2), 18. DOI
10. Giofrè, D., Toffalini, E., Perugini, A., **Esposito, L.**, Amoretti, G., & Geary, D. C. (2024). Sex differences in cognition: A meta-analysis of variance ratios in the Wechsler Intelligence Scales for Children. *Personality and Individual Differences*, 229, 112776. DOI
11. Giofrè, D., Toffalini, E., **Esposito, L.**, & Cornoldi, C. (2024). Sex/gender differences in general cognitive abilities: An investigation using the Leiter-3. *Cognitive Processing*, 25, 663–672. DOI
12. Giofrè, D., Toffalini, E., **Esposito, L.**, & Cornoldi, C. (2023). Gender differences in the Wechsler Intelligence Scale for Children in a large group of Italian children with attention deficit hyperactivity disorder. *Journal of Intelligence*, 11(9), 178. DOI

Manuscripts under revision

Masi, A., **Esposito, L.**, Cardillo, R., & Caviola, S. Math anxiety in secondary school children: The role of self-criticism and emotion regulation strategies.

Coppola, I., Rania, N., **Esposito, L.**, Puzzo, A. V., Giannoni, P., & Lagomarsino, F. Validation of the Intercultural Sensitivity Scale in Italian professional contexts.

Tironi, M., **Esposito, L.**, Giofrè, D., Smiley, P. A., Kerr, M. L., & Borelli, J. L. The Relational Savoring Questionnaire for Parents (RSQ-P): Development and validity among parents of young and school-aged children.

CONFERENCE PRESENTATIONS (SELECTED)

Esposito, L., & Giofrè, D. (2026, July 13–16). *Sex and group differences in intelligence: A meta-analysis of WISC profiles across typical and atypical development* [Poster]. International Society for Intelligence Research (ISIR), Warsaw, Poland.

Esposito, L., & Giofrè, D. (2025, July 14–18). *Investigating the impact of attitudes and learning environment on math achievement: Insights from the TIMSS data* [Oral presentation]. International Society for the Study of Individual Differences (ISSID), Vienna, Austria.

Esposito, L., Toffalini, E., & Giofrè, D. (2024, July 29–31). *Sex differences in intelligence on the WISC: A meta-analysis on children with specific learning disabilities* [Poster]. International Society for Intelligence Research (ISIR), Zurich, Switzerland.

Esposito, L., Toffalini, E., & Giofrè, D. (2024, March 1–2). *Una metanalisi sulle differenze di genere nei profili cognitivi di bambini e ragazzi con ADHD* [Poster]. XX Congresso Nazionale “Giornate di aggiornamento sull’uso degli strumenti in Psicologia Clinica dello Sviluppo”, Bologna, Italy.

Esposito, L., Masi, A., Bignetti, R., & Caviola, S. (2022, September 23–24). *Aspetti dominio-specifici e dominio-generalisti in compiti algebrici nei bambini della scuola secondaria* [Poster]. XXX Congresso Nazionale AIRIPA, Padua, Italy.

AWARDS AND GRANTS

Travel Grant Award, Institute of Mental Chronometry, for ISIR 2026 in Warsaw	2026
Travel Grant Award, Institute of Mental Chronometry, for ISIR 2024 in Zurich	2024

PEER REVIEW

Ad hoc reviewer for <i>Contemporary Educational Psychology</i>	2026
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METHODS, TEACHING AND OPEN MATERIALS

Methodological skills

Statistics. Structural equation and latent variable modelling (lavaan, semTools, sirt); meta-analysis (metafor, metaSEM, emmeans); multilevel models (lme4, lmerTest); item response theory (mirt, eRm).

Experiments. Design and implementation of behavioural experiments; programming in PsychoPy/Python; basic Psychtoolbox/MATLAB.

Software. R (proficient), Python (intermediate), MATLAB; RStudio, Mplus, JASP, LaTeX, Excel; Windows, macOS, Linux.

Reproducibility. R Markdown and Quarto; version control and sharing with GitHub and OSF; full data workflow from cleaning to preprocessing.

Teaching materials

 [GitHub repository](#)

Slides with reproducible code:

Introduction to meta-analysis, for PhD students

Scale validation and confirmatory factor analysis, research meeting

Introduction to mediation models, research meeting

Simulations

 [GitHub repository](#)

Basic simulations:

Meta-analytic SEM: convergence, bias and power of TSSEM vs OSMASEM

Class enumeration bias in latent profile analysis vs *k*-means clustering

Power of generalized linear mixed models vs random-effects meta-analysis in clustered data

Asymmetric outliers and inflated Type I error in interaction models

Additional training

Path analysis models for cross-sectional and longitudinal data, summer school Jun 2024

University “G. d’Annunzio” of Chieti-Pescara and Italian Psychological Association

Introduction to meta-analysis Apr 2024

Department of Psychology, Università Cattolica del Sacro Cuore (FORME MERITEA)

Electrophysiological signal analysis at different scales of complexity Dec 2023

University of Genoa

Tools for open science: Making your scientific research accessible and reproducible May 2023

Department of Developmental Psychology and Socialisation, University of Padua

EEGLAB: Basic analysis of electroencephalographic signals Nov 2022

Department of Developmental Psychology and Socialisation, University of Padua

Introduction to R: Basic elements and advanced applications Nov 2022

Department of Developmental Psychology and Socialisation, University of Padua

PsychoPy3: Programming an experiment Oct 2022

Department of Developmental Psychology and Socialisation, University of Padua

RMarkdown: Reproducible analyses, presentations, reports, and more in a single file Jun 2022

Department of Developmental Psychology and Socialisation, University of Padua

LANGUAGES

Italian (native)

English (B2, IELTS 2019: Listening 6.0, Reading 7.5, Writing 5.5, Speaking 6.0)